

Data Engineering Lab Midsem test

Date: 12.09.25

Time: 3 Hours

You are given a dataset for classification with m features, n samples and the target attribute (k possible values). Assume that all the features are discrete. You need to write python codes for the following three problems.

1. Fill out missing values of features by taking the most likely value of the feature for the class.
2. Rank the features based on their potential values (in descending order). The potential value a feature A is computed as

$$Potential(A) = R(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \times R(S_v)$$

where $R(S) = - \sum_{i=1}^c p_i \log_2 p_i$, p_i denotes the probability of a sample in S to be in class i and c denotes the total number of classes in the data. S_v denotes the set of samples in S that has value v for the attribute A.

3. Find out the most potential pair of features based on the following measure:

$$Potential(A, B) = R(S) - \sum_{(u,v) \in (Values(A) \times Values(B))} \frac{|S_{u,v}|}{|S|} \times R(S_{u,v})$$

*Submit this as a single python notebook file, with three clearly defined sections for the three problems. The name of your notebook file should be: **midsem-roll.ipynb**. A sample data is given for developing your program, where '-' means a missing value and the first row gives you the feature names.*

Print your output on screen in the following format.

Problem 1: sample id, feature name, value (one line for one record)

Problem 2: The feature name (one line for one feature)

Problem 3: Feature A, Feature B